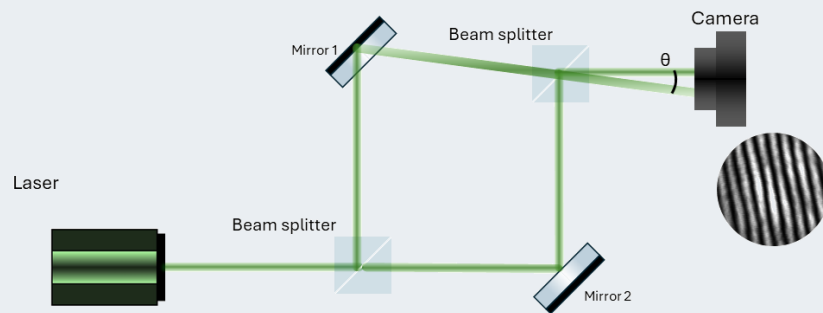


Optical Interferometers and Signal Detection

1 Off-axis digital holography

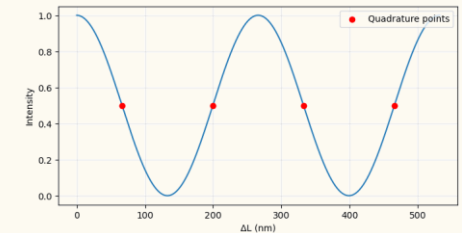
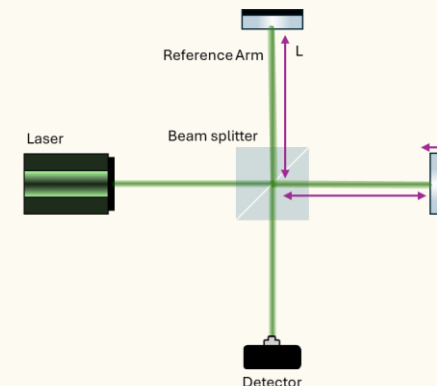
Mach-Zehnder Interferometer



- A **small tilt** between the reference and object beams creates off-axis interference fringes on the camera.
- The fringe pattern encodes the object wavefront, enabling digital reconstruction of the **spatial phase distribution**.

2 Active homodyne detection

Michelson Interferometer



- Environmental perturbations cause **phase drift**, moving the interferometer away from its optimal operating point.
- Active homodyne detection **locks** the interferometer at quadrature using **feedback**, giving a linear and sensitive phase readout.

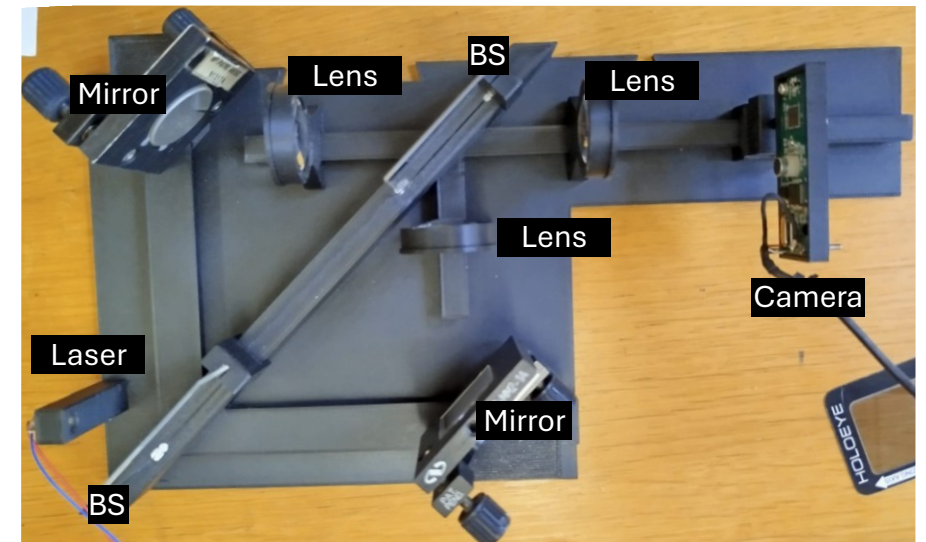
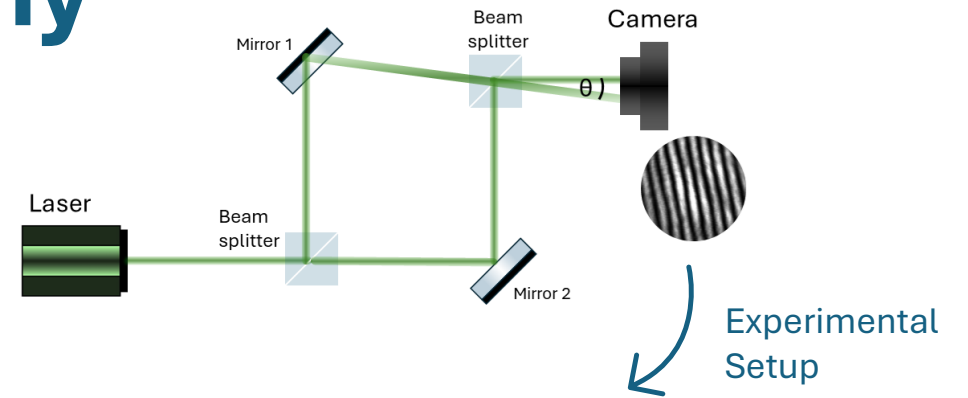
1. Off-axis digital holography

Activity 1 – Align the interferometer

Goal: obtain a stable off-axis fringe pattern on the camera.

What to observe:

- Beam overlap is required to generate interference
- Fringe visibility reflects alignment quality
- Fringe spacing/orientation depends on the reference beam tilt
- Small adjustments produce large changes in the pattern



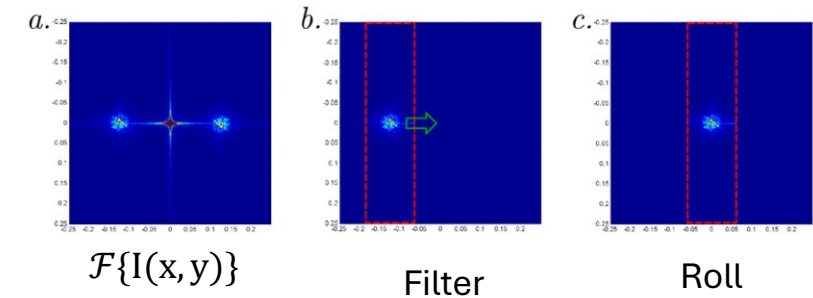
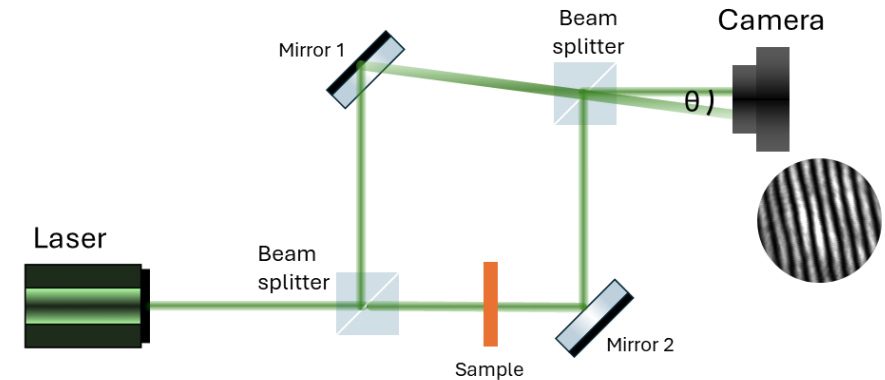
1. Off-axis digital holography

Activity 2 – Phase imaging of a transparent sample

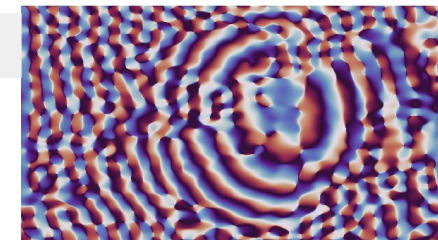
Goal: place a transparent sample in the object arm and reconstruct its phase profile.

What to observe:

- Local deformation of off-axis fringes.
- Fourier filtering isolates the object field.
- Reconstructed phase reveals thickness or refractive-index variations.



Reconstruct



1. Off-axis digital holography

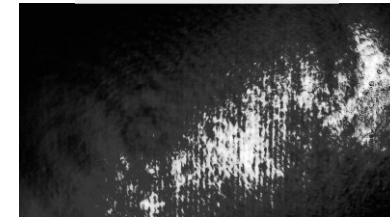
Activity 3 – Phase imaging of a transparent sample (with previous data)

Goal: reconstruct the phase profile of a transparent sample placed on the object path.

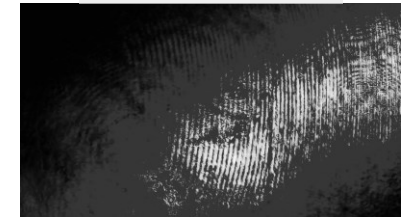
What to observe:

- Local deformation of off-axis fringes.
- Fourier filtering isolates the object field.
- Reconstructed phase reveals thickness or refractive-index variations.

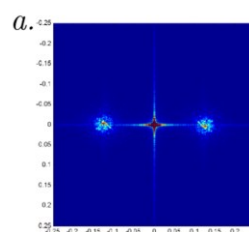
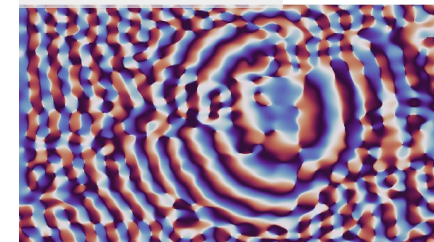
Reference



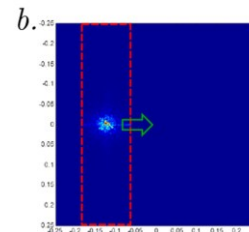
Interferogram



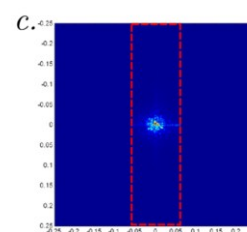
Reconstruct



$\mathcal{F}\{I(x,y)\}$



Filter



Roll



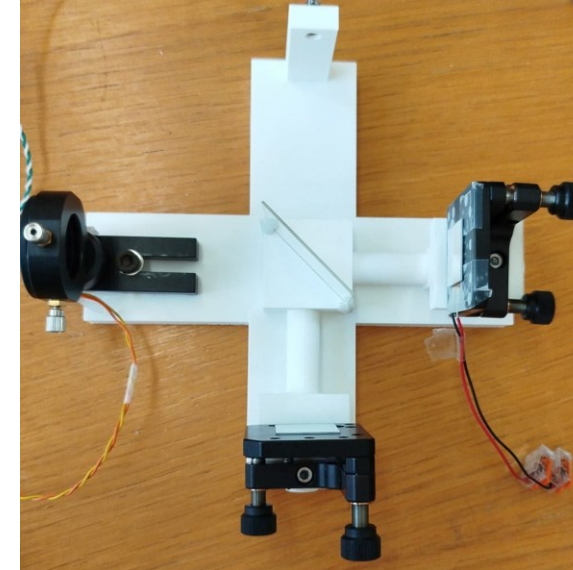
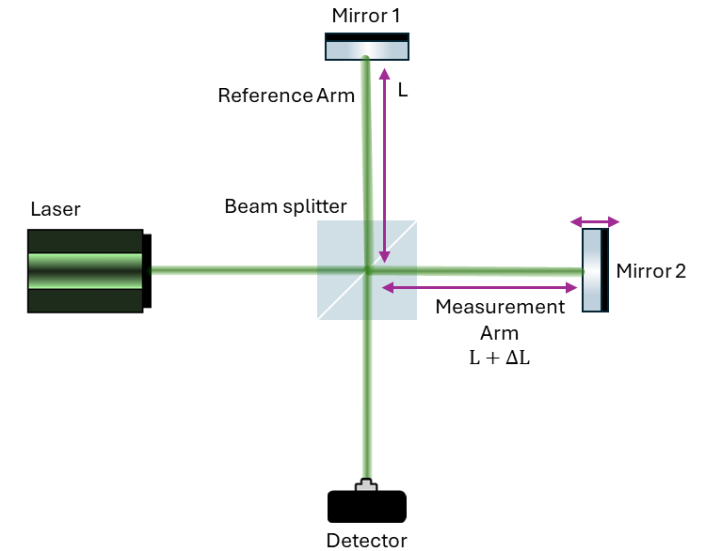
2. Active homodyne detection

Activity 1 – Align and listen to the interferometer

Goal: hear the effects of external perturbations on the interference.

What to observe:

- Mechanical perturbations produce audible changes.
- The response depends on the operating point of the interferometer.



2. Active homodyne detection

Activity 2 – Active homodyne locking

Goal: use feedback to keep the Michelson interferometer at quadrature.

What to observe:

Without feedback, the signal drifts due to environmental perturbations.

Feedback compensates slow phase drift.

The locked interferometer remains near maximum sensitivity.

Small phase variations are converted into a stable linear voltage signal.

